

[Science Bulletin] AI Model Predicts Internal Defects in Metal 3D-Printed Parts

Byeongku Lee 2026.01.15 17:00 KST



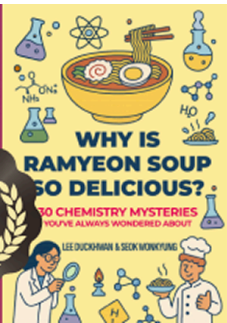
The 'Hanok, a House Built with Nature and Science' exhibition corner at the Gwacheon National Science Museum. Courtesy of Gwacheon National Science Museum

■ The Gwacheon National Science Museum announced on the 15th that it has fully renovated the 'Hanok and Traditional Korean Medicine' corner within its Korean Science and Civilization Hall, officially unveiling an exhibition where visitors can experience and understand the traditional scientific technologies of their ancestors. The renewal focuses on highlighting Hanok (traditional Korean houses) and traditional Korean medicine not merely as traditional culture, but as 'science in daily life' that observed nature and read changes in the body. The Hanok exhibition, themed 'Hanok, a House Built with Nature and Science,' presents the scientific wisdom of ancestors in managing wind, sunlight, and heat through interactive content. The traditional Korean medicine corner explores how traditional medicine interpreted bodily changes, focusing on pulse diagnosis and acupressure points, from the perspective of modern science. An integrated science interpretation program will be operated on weekends and

holidays from January 10 to February 22, 2026. "Through this exhibition renewal, visitors will be able to experience and understand Hanok and traditional Korean medicine through science and wisdom," said Han Hyeong-ju, Director General of the Gwacheon National Science Museum.

■ The Ministry of Science and ICT (MSIT) announced that Bae Kyung-hoon, Deputy Prime Minister and Minister of Science and ICT, and Kim Sung-hwan, Minister of Climate, Energy, and Environment, held the 'MSIT-Ministry of Climate Energy Policy Ondamhoe' at the Government Complex Seoul on the 15th. The term Ondamhoe signifies a meeting for warm and gentle communication. This meeting was organized to enhance policy coherence between ministries related to energy policy and to establish a practical cooperation framework in the future energy sector, in line with the push to accelerate the energy transition and strengthen the response to the climate crisis. The two ministries held in-depth discussions on major policy tasks in the future energy sector, including power supply for artificial intelligence (AI) data centers, commercialization of small modular reactors (SMRs), early realization of fusion energy, and carbon-neutral technology development. They plan to build practical cooperation models for matters requiring collaboration.

■ The Korea Institute of Materials Science (KIMS) announced on the 15th that a team led by Senior Researcher Park Jeong-min of the Nanomaterials Research Division, in a joint study with Germany's Max Planck Institute, has developed an AI model that can predict internal defects in metal 3D-printed parts in advance. The research results were published on January 1 (local time) in the international academic journal 'Acta Materialia.' The research team developed an 'explainable AI model' that can systematically analyze and predict the relationship between process conditions, defect formation, and mechanical performance during metal 3D printing. This approach allows for quality control by predicting the possibility of internal defect occurrence and performance changes from the process design stage. "This will contribute to increasing the industrial applicability of metal 3D printing in the manufacturing of high-performance components for fields such as aerospace and defense," said Senior Researcher Park.



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■ The Korea Industrial Technology Association (KOITA) announced that it held the '81st KOITA Breakfast Seminar' at the El Tower in Yangjae-dong, Seoul, on the 15th, featuring a lecture on the 2026 global economic outlook and U.S. monetary policy. The seminar was organized to explore strategic responses for companies amidst the rapidly changing global economic environment and to support the sustainable growth of its members. The lecture by Professor Kim Jin-il of Korea University, a former senior economist at the U.S. Federal Reserve Board (FRB, the Fed), presented an analysis of the world economic outlook from the International Monetary Fund (IMF) and domestic research institutes since the COVID-19 pandemic, as well as the role of U.S. monetary policy in the global forecast.

■ The Korean Academy of Science and Technology (KAST) announced on the 15th that it will hold the 247th KAST Roundtable on the morning of the 16th, both online and offline, under the theme 'What is R&D Failure?' to discuss the concept, policies, and cultural issues surrounding R&D failure from multiple perspectives. At the roundtable, Professor Stuart Firestein of Columbia University in the U.S. will give a special lecture on 'Failure: Why Science Is So Successful,' shedding light on the essential meaning of failure in science. The discussion, moderated by Professor Hong Sung-wook of Seoul National University, will feature panelists including Professor Lee Jeong-dong of Seoul National University, CEO Kim Myeong-gi of LSK Investment, and Team Leader Kim Min-soo of the Donga Science Daily News Team, who will discuss various issues such as evaluation and learning structures, commercialization links, and methods of social communication. The roundtable can be viewed online on the KAST's official YouTube channel (youtube.com/c/한국과학기술한림원1994).

<Reference>

- doi.org/10.1016/j.actamat.2025.121751

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